

2.5 Worksheet

1. Entry #95 in the table of integrals states $\int \frac{u^2}{\sqrt{u^2 + a^2}} du = \frac{u}{2} \sqrt{u^2 + a^2} - \frac{a^2}{2} \ln(u + \sqrt{u^2 + a^2}) + C$. Use this fact to determine the integral $\int \frac{2x^2}{\sqrt{2x^2 + 4}} dx$.

If we let $a = 2$, $u = \sqrt{2}x$, and $du = \sqrt{2}dx \Leftrightarrow \frac{1}{\sqrt{2}} du = dx$ then we have:

$$\begin{aligned} \int \frac{2x^2}{\sqrt{2x^2 + 4}} dx &= \frac{1}{\sqrt{2}} \int \frac{u^2}{\sqrt{u^2 + a^2}} du = \frac{1}{\sqrt{2}} \left[\frac{u}{2} \sqrt{u^2 + a^2} - \frac{a^2}{2} \ln(u + \sqrt{u^2 + a^2}) \right] + C \\ &= \frac{1}{\sqrt{2}} \left[\frac{\sqrt{2}x}{2} \sqrt{2x^2 + 4} - \frac{4}{2} \ln(\sqrt{2}x + \sqrt{2x^2 + 4}) \right] + C \end{aligned}$$

2. Entry #117 in the table of integrals states $\int \frac{du}{u\sqrt{2au - u^2}} = -\frac{\sqrt{2au - u^2}}{au} + C$. Use this fact to determine the integral $\int \frac{dx}{x\sqrt{6x - x^2}}$.

If we let $a = 3$, $u = x$, and $du = dx$ then we have:

$$\int \frac{dx}{x\sqrt{6x - x^2}} = \int \frac{du}{u\sqrt{2au - u^2}} = -\frac{\sqrt{2au - u^2}}{au} + C = -\frac{\sqrt{6x - x^2}}{3x} + C$$

3. Determine an entry in the table of integrals that can be used to evaluate the integral: $\int \frac{2x}{3x+4} dx$

Entry #60 in the table of integrals states $\int \frac{u du}{a+bu} = \frac{1}{b^2} [bu - a \ln|a+bu|] + C$.

If we let $a = 4$, $b = \frac{3}{2}$, $u = 2x$, and $du = 2dx \Leftrightarrow \frac{1}{2} du = dx$ then we have:

$$\int \frac{2x}{3x+4} dx = \frac{1}{2} \int \frac{u du}{bu+a} = \frac{1}{2} \left(\frac{1}{b^2} [bu - a \ln|a+bu|] \right) + C = \frac{1}{2} \left(\frac{4}{9} [3x - 4 \ln|4+3x|] \right) + C$$

4. Determine an entry in the table of integrals that can be used to evaluate the integral: $\int \frac{\ln(x)}{\sqrt{x^3}} dx$.

First note $\int \frac{\ln(x)}{\sqrt{x^3}} dx = \int x^{-3/2} \ln(x) dx$

We will use entry #47.

Let $n = -\frac{3}{2}$ and $u = x$, then $du = dx$. Therefore,

$$\begin{aligned} \int \frac{\ln(x)}{\sqrt{x^3}} dx &= \int x^{-3/2} \ln(x) dx \\ &= \int u^n \ln(u) du \\ &= \frac{u^{n+1}}{(n+1)^2} [(n+1) \ln(u) - 1] + C \\ &= \frac{x^{-1/2}}{(-1/2)^2} \left[\left(-\frac{1}{2} \right) \ln(x) - 1 \right] + C \\ &= \frac{4}{\sqrt{x}} \left[-\frac{1}{2} \ln(x) - 1 \right] + C \end{aligned}$$

5. Determine an entry in the table of integrals that can be used to evaluate the integral: $\int \frac{x}{(4-5x)^2} dx$.

We will use entry #102.

Let $a = 4$, $b = -5$ and $u = x$, then $du = dx$. Therefore,

$$\begin{aligned} \int \frac{x}{(4-5x)^2} dx &= \int \frac{u}{(a+bu)^2} du \\ &= \frac{a}{b^2(a+bu)} + \frac{1}{b^2} \ln|a+bu| + C \\ &= \frac{4}{25(4-5x)} + \frac{1}{25} \ln|4-5x| + C \end{aligned}$$

6. Determine an entry in the table of integrals that can be used to evaluate the integral: $\int \frac{\sqrt{3x^2-5}}{3x^2} dx$.

We will use entry #80.

Let $a = \sqrt{5}$ and $u = \sqrt{3}x$, then $du = \sqrt{3}dx \rightarrow \frac{1}{\sqrt{3}} du = dx$. Therefore,

$$\begin{aligned} \int \frac{\sqrt{3x^2-5}}{3x^2} dx &= \frac{1}{\sqrt{3}} \int \frac{\sqrt{u^2-5}}{u^2} du \\ &= \frac{1}{\sqrt{3}} \left[-\frac{\sqrt{u^2-a^2}}{u} + \ln|u + \sqrt{u^2-a^2}| \right] + C \\ &= \frac{1}{\sqrt{3}} \left[-\frac{\sqrt{3x^2-5}}{\sqrt{3}x} + \ln|\sqrt{3}x + \sqrt{3x^2-5}| \right] + C \end{aligned}$$

7. Determine an entry in the table of integrals that can be used to evaluate the integral: $\int \frac{\cos(4x)}{9 + \sin^2(4x)} dx$.

We will use entry #16.

Let $a = 3$ and $u = \sin(4x)$, then $du = 4 \cos(4x) dx \rightarrow \frac{1}{4} du = \cos(4x) dx$. Therefore,

$$\begin{aligned} \int \frac{\cos(4x)}{9 + \sin^2(4x)} dx &= \frac{1}{4} \int \frac{1}{a^2 + u^2} du \\ &= \frac{1}{a} \tan^{-1}\left(\frac{u}{a}\right) + C \\ &= \frac{1}{3} \tan^{-1}\left(\frac{\sin(4x)}{3}\right) + C \end{aligned}$$

8. Determine an entry in the table of integrals that can be used to evaluate the integral: $\int x^3 \sqrt{5 - 9x^2} dx$.

First note that $\int x^3 \sqrt{5 - 9x^2} dx = \int x \cdot x^2 \sqrt{5 - 9x^2} dx$

We will use entry #105.

Let $a = 5$, $b = -9$ and $u = x^2$, then $du = 2x dx \rightarrow \frac{1}{2} du = x dx$. Therefore,

$$\begin{aligned} \int x^3 \sqrt{5 - 9x^2} dx &= \int x \cdot x^2 \sqrt{5 - 9x^2} dx \\ &= \frac{1}{2} \int u \sqrt{a + bu} du \\ &= \frac{1}{2} \left[\frac{2}{15b^2} (3bu - 2a)(a + bu)^{3/2} \right] + C \\ &= \frac{1}{2} \left[\frac{2}{15(81)} (-27x^2 - 10)(5 - 9x^2)^{3/2} \right] + C \end{aligned}$$